

Testudo

User Manual

A native, local-first macOS environment for structured professional and research work

Manual version:	2.3
Application version:	0.3.2 (build 18)
Platform:	macOS 26+, Apple Silicon
Date:	October 2026

Foivos Karakostas

About this manual

This manual documents the implemented behaviour of **Testudo** 0.3.2 (build 18). It is intentionally more than a feature list. It is written so that a new user can understand the data model, perform ordinary work, and actively verify that the application behaves as expected.

Each major chapter therefore contains practical procedures and **Try it** exercises. These exercises are safe to perform in the Demo Environment or in a disposable Work Environment. They are especially useful when learning the application after an update.

Version covered

This edition documents Testudo 0.3.2 (build 18). This maintenance release corrects archived Task visibility in active Task lists and Parent Task selection, while preserving the archived hierarchy and historical relationships. It also restores the detail panel for Discontinued Tasks. No additional database schema migration is introduced. Testudo 0.3.1 introduced optional, time-zone-aware Planned Start, inherited availability constraints for Subtasks, Calendar and Today integration, and historical change tracking. Environments using Planned Start transition to database schema 9. Older Testudo versions cannot open schema 9 Environments. Schema 8 remains supported; schema 7 requires explicit migration. Simultaneous multi-device editing and automatic conflict resolution are not supported.

Testudo is an early public release. Keep independent backups of important `.testudoenv` packages. When experimenting with unfamiliar operations, prefer the Demo Environment first.

Terminology

An **Activity** is a chronological Work item that records something that happened. A **Calendar Event** is a separate scheduled object. The two concepts are deliberately different even when they refer to the same real-world work.

Contents

About this manual i

1 What Testudo Is 1
- 1.1 Purpose . . . 1
- 1.2 The main domains . . . 1
- 1.3 Local-first architecture . . . 1

2 Requirements, Installation and Updating 3
- 2.1 Supported baseline . . . 3
- 2.2 Quick installation . . . 3
- 2.3 Compatibility check only . . . 3
- 2.4 Manual source build . . . 3
- 2.5 Updating . . . 4

3 First Launch and the Local Testudo Profile 5
- 3.1 Create or restore a profile . . . 5
- 3.2 Importing a portable profile . . . 5
- 3.3 Application password and recovery . . . 5

4 Local Log Out, Local Sign Out and Environment Sign Out 6
- 4.1 Log Out of Testudo . . . 6
- 4.2 Sign Out from Testudo . . . 6
- 4.3 Sign Out of Environment . . . 6

5 Work Environments 7
- 5.1 The Work Environments screen . . . 7
- 5.2 Creating an Environment . . . 7
- 5.3 Opening an existing Environment . . . 7
- 5.4 Environment login . . . 7
- 5.5 Forgetting an Environment from one Mac . . . 8

6 Environment Membership and Administration 9
- 6.1 Environment roles . . . 9
- 6.2 Membership identity . . . 9
- 6.3 Environment passwords . . . 9
- 6.4 Environment recovery questions . . . 9

7 The Three-Column Interface 10
- 7.1 Sidebar . . . 10
- 7.2 Middle column . . . 10
- 7.3 Detail pane . . . 10
- 7.4 Sidebar visibility . . . 10
- 7.5 Back and Forward navigation . . . 11
- 7.6 Printing the detail pane . . . 11
- 7.7 Exporting detail content – Beta . . . 11

8 Work: Tasks, Notes and Activities 13
- 8.1 Three Work types . . . 13
- 8.2 Creating Work . . . 13
- 8.3 Editing existing Work . . . 13
- 8.4 Creating child Work quickly . . . 13

9 Task Workflow Status **15**
9.1 The four statuses . . . 15
9.2 To Do . . . 15
9.3 In Progress . . . 15
9.4 Completed . . . 15
9.5 Changing status from the detail view . . . 15

10 Discontinued Tasks **16**
10.1 Why Discontinued exists . . . 16
10.2 Required metadata . . . 16
10.3 What remains visible . . . 16
10.4 Historical cycles . . . 16

11 Closing, Archive and Reopening **17**
11.1 Closed is not a status . . . 17
11.2 What closing does . . . 17
11.3 Archive presentation . . . 17
11.4 Reopen . . . 17
11.5 Backward compatibility . . . 17

12 Notes **19**

13 Activities and the Work Log **20**
13.1 Activity semantics . . . 20
13.2 Activity in Timeline and History . . . 20

14 Themes and Multi-Theme Classification **21**
14.1 Purpose . . . 21
14.2 Theme detail fields . . . 21
14.3 Multiple Themes on one Work item . . . 21
14.4 Descendant aggregation . . . 21

15 People **22**
15.1 Person profile fields . . . 22
15.2 Affiliations . . . 22
15.3 Environment membership link . . . 22

16 Organizations and Groups **23**
16.1 Organizations . . . 23
16.2 Groups . . . 23
16.3 Affiliated People . . . 23

17 Work Relationships **24**
17.1 Inheritance by child items . . . 24

18 Related Work **25**

19 Planned Start **26**
19.1 Purpose . . . 26
19.2 Subtasks . . . 26
19.3 Today and Calendar . . . 26
19.4 History . . . 26
19.5 Compatibility . . . 26

20 Today **27**
20.1 Today is a work summary, not only a schedule . . . 27
20.2 Needs Attention . . . 27
20.3 Upcoming . . . 27

21 Task Lists and Timeline **28**
21.1 Timeline . . . 28

22 History and the Work Log **29**
22.1 Audit timestamps . . . 29
22.2 Log events . . . 29
22.3 Deletion and history . . . 29

23 Calendar Overview **30**
23.1 What appears in Calendar . . . 30
23.2 Day workspace . . . 30
23.2.1 Planned Work . . . 30
23.2.2 Day History . . . 30
23.2.3 App Log . . . 31
23.3 Date navigation . . . 31
23.4 Calendar sources . . . 31

24 Creating and Editing Calendar Events **32**
24.1 New Calendar Event . . . 32
24.2 Calendar Event status . . . 32
24.3 External events . . . 32

25 Apple Calendar Integration **33**
25.1 Connecting Apple Calendar . . . 33
25.2 Permission and availability . . . 33

26 iCal Subscriptions **34**

27 Time Zones and Historical Dates **35**
27.1 Absolute time and source zone . . . 35
27.2 Historical entry . . . 35

28 Demo Environment **36**
28.1 Purpose . . . 36
28.2 Demo naming . . . 36
28.3 Suggested tour . . . 36

29 Storage, Portability and Backups **37**
29.1 Portable Environment package . . . 37
29.2 Storage compatibility in Testudo 0.3.0 . . . 37
29.3 Safer startup and Environment switching . . . 37
29.4 Environment credential-file safety . . . 37
29.5 Administrator session requirement . . . 38
29.6 Cloud-synchronised folders . . . 38
29.7 Backups . . . 38

30 Security and Privacy **39**
30.1 Passwords . . . 39
30.2 Security questions . . . 39
30.3 External calendar secrets . . . 39
30.4 Sensitive portable files . . . 39

31 Deletion Semantics **40**

32 Recommended Working Practices **41**
32.1 Separate plan, context and evidence . . . 41
32.2 Use Discontinued instead of pretending success . . . 41
32.3 Close only when the top-level work is no longer part of active workflow . . . 41
32.4 Use Themes for concepts, not as a duplicate Task tree . . . 41
32.5 Model recurring professional context explicitly . . . 41
32.6 Record outcomes as Activities . . . 41

33 Troubleshooting **42**
33.1 An Environment requires migration . . . 42

33.2 Credentials are missing or incompatible . . . 42
33.3 An Administrator cannot change a password . . . 42
33.4 The installer says the source checkout has local changes . . . 42
33.5 Apple developer tools are missing . . . 42
33.6 An Environment package was moved . . . 43
33.7 Environment login fails . . . 43
33.8 Apple Calendar does not appear . . . 43
33.9 iCal synchronisation fails . . . 43
33.10 A Task cannot be Closed . . . 43
33.11 A Discontinued status change appears unavailable in a shortcut . . . 43
33.12 Archived Work is missing from active lists . . . 43

34 Guided End-to-End Verification Lab **44**
34.1 Step 1: create the Environment . . . 44
34.2 Step 2: create professional context . . . 44
34.3 Step 3: create a top-level Task . . . 44
34.4 Step 4: create children . . . 44
34.5 Step 5: exercise status and Today . . . 45
34.6 Step 6: exercise Discontinued . . . 45
34.7 Step 7: exercise Completed and Archive . . . 45
34.8 Step 8: navigation history . . . 45
34.9 Step 9: Reopen . . . 45
34.10 Step 10: calendar . . . 45

35 Technical Reference **46**
35.1 Important file extensions . . . 46
35.2 Source tree . . . 46
35.3 Build scripts . . . 46
35.4 Version identity . . . 46
35.5 Task status persistence compatibility . . . 47

36 Glossary **48**

A Command Reference **49**
A.1 Check compatibility . . . 49
A.2 Install . . . 49
A.3 Clone manually . . . 49
A.4 Update source . . . 49
A.5 Build local installer . . . 49
A.6 Build application only . . . 49
A.7 Request Apple Command Line Tools . . . 49

B Manual Source and Compilation **50**

C Quick Acceptance Checklist **51**

D Early Release Notice **52**

License **53**

Author **54**

CHAPTER 1

What Testudo Is

1.1 Purpose

Testudo is a native macOS application for work that is difficult to represent with a flat task list. It combines actionable work, durable context, a chronological work log, professional entities, semantic classification, calendars, and local portable storage.

Typical use cases include:

- scientific research and laboratory work;
- research software engineering;
- technical and infrastructure operations;
- academic collaborations and committees;
- long-running projects with many sub-problems;
- work involving multiple institutions, teams or people;
- activity where preserving the history of decisions and outcomes is important.

The core design principle is that **structure** and **chronology** are both first-class. A Task can have children and Themes, but Testudo also preserves when the work started, completed, was discontinued, was closed, or was reopened.

1.2 The main domains

Testudo connects several domains that are often kept in separate applications:

Domain	Purpose
Work	Tasks, Notes and Activities, including unlimited Task nesting.
Themes	Hierarchical semantic classification, independent from Task hierarchy.
Entities	People, Groups and Organizations describing professional context.
Relationships	Semantic links such as For, Requested by, With, Assigned to and Related to.
Calendar	Testudo events plus read-only external calendar sources.
History	Significant lifecycle and relationship events.
Work Environments	Portable, independent `.testudoenv` packages.
Identity	A local Testudo profile plus separate Environment memberships.

1.3 Local-first architecture

Testudo does not require a Testudo cloud service. Work data lives in the selected `.testudoenv` package. The application also stores installation-level information such as the local user profile, registered Environment locations, filesystem bookmarks and remembered sign-ins.

This separation matters:

- copying a `.testudoenv` package copies the Work Environment;

- signing out from the local Testudo installation does not delete the Environment package;
- forgetting an Environment from one Mac does not delete its package;
- a cloud-synchronised folder can transport the package, but Testudo is still operating on files rather than a central multi-user server.

After opening Testudo, identify: (1) your local Testudo user, (2) the active Work Environment, and (3) your Environment membership. These are related but separate identities.

CHAPTER 2

Requirements, Installation and Updating

2.1 Supported baseline

The current source targets:

- macOS 26 or later;
- Apple Silicon;
- an Apple developer toolchain with Swift 6 or later;
- Git.

The repository uses a source-first installation model. The application is compiled on the Mac where it is used and is ad-hoc signed by the local build script.

2.2 Quick installation

Run:

```
curl -fsSL \
  https://raw.githubusercontent.com/foivoskar/Testudo/main/install.sh \
  | bash
```

The installer checks the machine, clones or updates the repository, refuses to overwrite a checkout with local modifications, builds Testudo, creates a DMG, verifies it and opens the finished installer.

When the DMG opens, drag `Testudo.app` into `/Applications`.

2.3 Compatibility check only

To test the toolchain without cloning or building:

```
curl -fsSL \
  https://raw.githubusercontent.com/foivoskar/Testudo/main/install.sh \
  | bash -s -- --check
```

2.4 Manual source build

```
git clone https://github.com/foivoskar/Testudo.git
cd Testudo
./Scripts/build-local-dmg.sh
```

The resulting DMG is written under `dist/`. For this release the expected name is normally:

```
dist/Testudo-0.3.2-macOS-arm64.dmg
```

2.5 Updating

If the source checkout is clean:

```
cd ~/Testudo
git pull --ff-only
./Scripts/build-local-dmg.sh
```

Replace the previous copy in `/Applications` with the newly built application.

Build and install Testudo, then confirm that the application identifies itself as 0.3.2 build 18. If you maintain a development checkout, also inspect `Scripts/version.sh` and confirm that it contains the same version and build number.

CHAPTER 3

First Launch and the Local Testudo Profile

3.1 Create or restore a profile

On first use, Testudo presents **Create or Restore Your Profile** . The local profile belongs to this installation, not to a Work Environment.

The setup interface can record professional information including:

- first, middle, last and preferred name;
- academic title and job title;
- professional email, phone and office;
- website, ORCID, LinkedIn and GitHub;
- professional address, city, postal code and country;
- professional fields, responsibilities and notes.

Only enter fields that are useful to you. The profile can later be edited from the application settings.

3.2 Importing a portable profile

If you previously exported a `.testudouser` file, you can import it instead of re-entering profile data.

A `.testudouser` file restores profile information and avatar data. It does *not* restore:

- the application password;
- registered Work Environment locations;
- Environment passwords;
- remembered Environment sign-ins.

3.3 Application password and recovery

After creating the profile, Testudo can protect the installation with an application password. The setup explicitly states that this password is separate from every Work Environment password.

The application password must contain at least eight characters. Recovery uses three user-defined security questions.

Security-question answers are stored as salted hashes. Matching normalises capitalization, accents and repeated spaces.

> Create a disposable local profile, configure the application password and three recovery questions, use **Log Out of Testudo** , and log back in. Confirm that your registered Environment list is still present. Then compare this with **Sign Out from Testudo…** , which is intentionally much more destructive to local installation state.

CHAPTER 4

Local Log Out, Local Sign Out and Environment Sign Out

Testudo exposes several actions that sound similar but have different scopes.

4.1 Log Out of Testudo

Logging out locks the application. Your local profile, registered Work Environments and remembered state remain available after the next login.

Use this when you simply want to protect the application while keeping the current installation configured.

4.2 Sign Out from Testudo

This is an installation-level reset for the local Testudo identity. The confirmation screen requires typing `SIGN OUT`.

It removes local information from this Mac, including the local profile, application password, registered Work Environments, local bookmarks, Environment identity mappings and remembered sign-ins.

It does **not** delete or modify the actual `.testudoenv` packages.

Before signing out, Testudo offers **Export My Profile Before Signing Out...** .

4.3 Sign Out of Environment

This action affects the active Work Environment session rather than the local application account. It is useful when the same local Testudo installation is used to enter an Environment with different credentials.

> If you only want to lock the application, use **Log Out of Testudo** . If you want to remove local Testudo-user state from the Mac, use **Sign Out from Testudo...** . If you only want to end the current Environment session, use **Sign Out of Environment** .

CHAPTER 5

Work Environments

5.1 The Work Environments screen

After local authentication, Testudo shows **Work Environments** with the message *Choose where you want to work*. Registered Environments appear as cards showing their name and storage path.

Available actions are:

- **Create New Environment** ;
- **Open Existing Environment...** ;
- **Load Demo Environment** .

5.2 Creating an Environment

The creation sheet asks for:

- Environment name;
- initial Administrator username;
- Administrator password;
- password confirmation.

The first account belongs to the Work Environment and is separate from the local Testudo profile.

The Environment is saved as a portable `.testudoenv` package. You can choose a local, cloud-synchronised or external storage location.

Internally, a normal package contains files such as:

```
Example.testudoenv/
|-- EnvironmentManifest.json
|-- EnvironmentData.json
|-- EnvironmentCredentials.json
‘-- EnvironmentIcon.png        # optional
```

5.3 Opening an existing Environment

Use **Open Existing Environment...** and select a `.testudoenv` package. Testudo validates the package and registers its location with the current installation.

If the Environment requires identity selection, Testudo presents the Environment login flow.

5.4 Environment login

The entry sheet contains a username, password and **Stay signed in** option.

When **Stay signed in** is enabled, this Mac can reopen the Environment without asking for those credentials until the user explicitly signs out of the Environment.

5.5 Forgetting an Environment from one Mac

In Environment settings, removing an Environment from Testudo only forgets its registration on that installation. The `.testudoenv` package and all Environment data remain at the storage location and can be reopened later.

Create a disposable Environment called *Manual Test*. Close it, return to Work Environments, and reopen it. Then remove its registration from Testudo and use **Open Existing Environment...** to register the same package again. Verify that the Work data is still present.

CHAPTER 6

Environment Membership and Administration

6.1 Environment roles

Environment memberships currently have two roles:

- Administrator;
- User.

A membership also has an active/inactive state.

6.2 Membership identity

An Environment membership can either be linked to an existing Person entity or maintain its own first and last name. When linked to a Person, the membership name follows the Person.

The management view exposes:

- Linked Person;
- first and last name when unlinked;
- username;
- immutable Membership ID;
- Environment role;
- active membership state.

6.3 Environment passwords

Environment passwords are separate from the local application password.

An Administrator can assign a new Environment password without knowing the member's current password. Security-question answers are never revealed to Administrators.

6.4 Environment recovery questions

A member can configure three recovery questions for the current Environment account. These recovery questions belong to that Environment account and travel with the `.testudoenv` package.

If a member cannot answer a recovery question, an Environment Administrator can still assign a new password.

> In a disposable Environment, create or inspect a second membership. Confirm that changing the Environment password does not change the Testudo application password. If the membership is linked to a Person, edit the Person name and verify that the membership display follows the linked Person.

CHAPTER 7

The Three-Column Interface

7.1 Sidebar

The left sidebar is the global navigation layer. The current sidebar contains:

- New Entry;
- Today;
- Calendar;
- All Tasks;
- To Do;
- In Progress;
- Completed;
- Discontinued;
- Archive;
- Timeline;
- Themes;
- Organizations;
- Groups;
- People.

7.2 Middle column

The middle column shows the collection selected in the sidebar: a Task list, Archive tree, calendar, Theme list, entity list or other contextual collection.

Task-oriented middle-column views prioritise Tasks with deadlines. Earlier deadlines appear first, followed by Tasks without deadlines. In **All Tasks** , the same rule applies recursively among Sub-task siblings. **All Tasks** represents active Work only: once a terminal top-level Task is Closed, that Task and its complete descendant tree remain available through Archive but no longer appear in **All Tasks** . Active **Today** work uses the same deadline priority where Task ordering is relevant.

7.3 Detail pane

The right pane shows the selected object's detail view or a creation/editing form. Normal detail views are mostly read-only and expose a single whole-object **Edit** action where appropriate.

7.4 Sidebar visibility

Testudo uses the native macOS sidebar behaviour. The system-provided sidebar control hides and restores the first column.

7.5 Back and Forward navigation

The detail pane maintains browser-style navigation history through Back and Forward controls and two-finger trackpad navigation.

The navigation history represents the complete three-column context. When you move backward or forward, Testudo restores:

- the sidebar section;
- the middle-column selection;
- the right-pane detail content.

Archived Tasks participate in the same history as active Work.

> Open an active Task, then open **Archive** and select a Closed Task, then open a Person. Use Back and Forward or the trackpad gesture. Verify that each step restores not only the right pane but also the corresponding sidebar section and middle-column selection.

7.6 Printing the detail pane

Press Command-P to open the native macOS Print dialog for the current right-hand detail content.

Testudo prints the detail content rather than the navigation sidebar or middle-column collection. When the current detail view is scrollable, Testudo captures the complete scrollable content and exposes it to the print system as one continuous document. macOS can therefore paginate a long Task, Day, Person, Theme or other supported detail view across as many pages as required.

Printed detail content uses a 90 percent scale. Testudo otherwise retains the normal print-margin behaviour supplied by the macOS print system and selected printer.

> Open a Task or Day whose detail content requires substantial scrolling and press Command-P. Confirm that the native Print dialog opens, that the sidebar and middle column are absent, and that the complete detail content continues across multiple pages rather than stopping at the visible viewport.

7.7 Exporting detail content — Beta

Choose **File > Export...** while a supported detail destination is open.

The current Export implementation is a Beta feature. Its first output format is PDF.

Unlike detail-pane printing, PDF Export is generated semantically from Testudo data rather than by capturing the visible interface as an image. Text in the exported PDF therefore remains real selectable text and stays sharp when the document is enlarged.

The exporter reproduces important parts of Testudo's detail visual language, including:

- the main title and object type;
- relevant SF Symbols and status symbols;
- Theme and status context shown below Work titles;
- structured label/value detail rows;
- related and contained Work where applicable;
- lifecycle History fields.

The detailed Work Log is intentionally not included in exported Work documents.

For a Task that contains child Work, Export can optionally include nested items recursively. Enable **Include nested items** and choose the desired depth.

Depth 1 includes direct children. Larger values continue through deeper descendants. The maximum selectable value is calculated dynamically from the deepest branch that actually exists beneath the selected Task, so choosing **Full depth** includes the complete available subtree.

Export is functional but its document layout is still under development. It does not yet reproduce every visual detail of every Testudo pane exactly. The semantic structure, typography, symbols and supported metadata will continue to be refined in later releases.

Open a Task containing at least two levels of child Work. Choose **File > Export...** , enable nested items and select **Full depth** . Open the resulting PDF and confirm that the root Task and descendants are included as real selectable text and that no application scrollbars or Work Log section appear in the document.

CHAPTER 8

Work: Tasks, Notes and Activities

8.1 Three Work types

Task Actionable work with a Task status and optional lifecycle dates.

Note Durable context that does not require Task status.

Activity A work-log item recording something that happened, optionally at a specific date and time.

Only Tasks can contain child Work. A Task may contain sub-tasks, Notes and Activities, and sub-tasks can contain further Work without a fixed nesting limit.

8.2 Creating Work

Choose **New Entry** or press Command-N. The Work editor includes sections for content, classification, parentage, timing and semantic relationships.

Common fields include:

- Type;
- one or more Themes;
- Parent Task;
- optional Title;
- Description or Note / What happened? text;
- relationships to People, Groups and Organizations.

At least a title or description is required.

8.3 Editing existing Work

Open the Work item and choose **Edit** . The same complete editor is used for Tasks, Notes and Activities. Existing values are pre-filled.

8.4 Creating child Work quickly

Inside a Task detail page, the **Contains** area can create:

- Sub-task;
- Note;
- Activity.

Activities created this way are still Work-log objects, not Calendar Events.

The **Contains** list is shown newest first. Activities are positioned by their **Occurred** time when one is available; Tasks and Notes use their Created time. Deadline does not control this sequence because **Contains** represents chronology rather than priority.

Create a top-level Task called *Prepare field campaign*. Add a sub-task called *Check instruments*, a Note with background information and an Activity recording a completed phone call. Then add a deeper sub-task under *Check instruments*. Navigate up and down the hierarchy and confirm that each item retains its own detail page.

CHAPTER 9

Task Workflow Status

9.1 The four statuses

In Testudo 0.1.7, the user-facing Task statuses are exactly:

- To Do;
- In Progress;
- Completed;
- Discontinued.

Closed is deliberately absent from this list because it is an archive lifecycle action rather than a workflow outcome.

9.2 To Do

Use To Do for work that exists but has not begun. A Task can still have a deadline while remaining To Do.

9.3 In Progress

Use In Progress when work has begun. When a Task changes from To Do to In Progress and does not already have a Started date, Testudo can record the transition time automatically. A manually entered Started time remains independently editable.

9.4 Completed

Use Completed when the work ended successfully. The completion timestamp is preserved independently from Created and Started.

9.5 Changing status from the detail view

The status control allows normal workflow changes. Discontinued requires additional metadata, so it is intentionally handled through the full Work editor rather than a shortcut that could omit the reason.

> Create a Task as To Do. Change it to In Progress and check its Started value. Then mark it Completed and confirm that a completion time appears. Reopen the editor and verify that Created, Started and Completed represent different concepts and can describe historical work accurately.

CHAPTER 10

Discontinued Tasks

10.1 Why Discontinued exists

Not every Task finishes successfully. Work may fail, become unnecessary, be replaced by another approach, or prove infeasible. Deleting such a Task loses useful history; marking it Completed is misleading. Testudo therefore provides a terminal **Discontinued** status.

10.2 Required metadata

When the Task status is Discontinued, the editor requires a reason. Available reasons are:

- Failed;
- Superseded;
- No longer needed;
- Not feasible;
- Other.

The editor also includes:

- Discontinued date/time with source time zone;
- optional Outcome / note.

The discontinued time cannot be in the future and cannot be earlier than Started when a Started time exists.

10.3 What remains visible

The normal detail page shows the Discontinued time, reason and optional outcome. The Task also appears in the **Discontinued** sidebar section while it remains active rather than archived. Selecting the Task in that section opens its detail page in the right pane, including the normal applicable editing and deletion controls.

A discontinuation lifecycle event is retained in History.

10.4 Historical cycles

If a Task later returns to active work, earlier discontinuation history remains preserved. A later discontinuation can therefore represent a new lifecycle cycle without erasing the previous one.

> Create a Task called *Use vendor API A*. Start it, edit it to Discontinued, choose *Superseded*, and write *Replaced by API B*. Confirm that the Task appears in **Discontinued** , that its detail page shows the reason and outcome, and that its Log contains a discontinuation event.

CHAPTER 11

Closing, Archive and Reopening

11.1 Closed is not a status

Closing is an archival lifecycle operation. Only a **top-level Task** can be Closed, and it must already be terminal:

- Completed; or
- Discontinued.

A To Do or In Progress Task cannot be closed directly.

11.2 What closing does

Closing a top-level Task:

- records a Closed timestamp and source time zone;
- retains the real workflow status;
- moves the whole Task subtree to Archive;
- records a Closed lifecycle History event;
- removes the subtree from active Work queries.

Children do not independently become Closed. They are archived because their top-level ancestor is closed.

11.3 Archive presentation

The Archive middle column displays Closed top-level Tasks and their retained subtree. The root still indicates whether the actual outcome was Completed or Discontinued, and its context shows the Closed time.

11.4 Reopen

A Closed top-level Task exposes **Reopen**. Reopening removes the archive marker, returns the tree to active views, preserves the actual Completed or Discontinued status and records a Reopened History event.

11.5 Backward compatibility

Testudo 0.1.6 stored Closed as a Task status. Testudo 0.1.7 keeps a legacy persistence-only value so older Environment data can be decoded, then migrates the item to the separate archive lifecycle while preserving the most meaningful previous workflow state available.

Perform the following in a disposable Environment:

1. Create a top-level Task and a child Note.
2. Mark the Task Completed.
3. Use **Close** in the detail header.
4. Confirm that the Task disappears from active lists and appears in **Archive** with its child Note.
5. Open the archived Task and verify that its Status is still Completed.
6. Use Back/Forward navigation and confirm Archive selection is restored correctly.
7. Choose **Reopen** .
8. Confirm that the Task returns to active views as Completed and that Closed/Reopened events are preserved in its Log.

Repeat the exercise with a Discontinued Task and verify that its discontinuation reason and outcome survive Close and Reopen.

CHAPTER 12

Notes

Notes are informational Work items. They do not have Task status and are useful for durable context that belongs near the work.

A Note can have:

- title and body;
- one or more Themes;
- a Parent Task;
- a Reminder;
- semantic relationships.

Use a Note when the item should remain available as context but does not represent work that progresses through To Do, In Progress, Completed or Discontinued.

> Inside a Task, create a Note containing a decision or reference, enable a Reminder for tomorrow, and save it. Confirm that it appears under the parent Task and that the reminder is represented in Calendar and relevant Today views when its date becomes current.

CHAPTER 13

Activities and the Work Log

13.1 Activity semantics

An Activity records something that happened while working. Examples include a discussion, measurement, intervention, review, decision or resolved problem.

The Activity editor provides an optional **Occurred** date/time with source time zone.

Activity vs Calendar Event

An Activity belongs to the Work history. A Calendar Event belongs to the Calendar domain. A meeting that is planned for tomorrow is a Calendar Event; after it occurs, you may also record an Activity containing the useful outcome.

13.2 Activity in Timeline and History

Activities contribute to chronological views and appear in the Work Log. A parent Task can therefore preserve a running record without rewriting its description repeatedly.

Create a Calendar Event for a meeting and link it to a Task. After the simulated meeting, create an Activity under the Task called *Meeting outcome* and record a historical Occurred time. Compare how the Calendar Event and Activity appear in their respective views.

CHAPTER 14

Themes and Multi-Theme Classification

14.1 Purpose

Themes classify Work semantically and independently from the Task hierarchy and professional entity hierarchy.

A Theme can have a parent Theme, allowing arbitrary Theme depth.

14.2 Theme detail fields

The current Theme interface exposes information including:

- Name and Parent Theme;
- Summary and Notes;
- Start and Target dates;
- Tags;
- URL;
- SF Symbol;
- related People, Groups and Organizations;
- Created and Updated timestamps.

The normal detail page is read-only and exposes a whole-object **Edit** action.

14.3 Multiple Themes on one Work item

A Work item can belong to several Themes simultaneously. This avoids duplicating the same Task simply because it belongs to multiple conceptual domains.

14.4 Descendant aggregation

Related Work for a parent Theme includes Work assigned to descendant Themes recursively. This aggregation does not rewrite direct membership.

> Create Theme A, then child Theme B. Create a Task assigned directly to Theme B and also to a second independent Theme. Open Theme B and Theme A and verify that the Task appears in Related Work for both levels of the A/B hierarchy, while the Task editor still shows its actual direct Theme selections.

CHAPTER 15

People

People are first-class professional entities with richer profile data than a simple name string.

15.1 Person profile fields

The Person editor can represent:

- first, middle, last and preferred name;
- academic title and position;
- primary and secondary email/phone;
- office and assistant/contact;
- staff or employee ID;
- address and workplace;
- website, LinkedIn and GitHub;
- ORCID, ResearcherID, Scopus Author ID and Google Scholar;
- fields/expertise, responsibilities and tags;
- preferred language and time zone;
- notes;
- Organization and Group affiliations.

15.2 Affiliations

A Person can belong to multiple Organizations and Groups. One affiliation can be primary. Some affiliations can be shown as automatic when implied by structure or membership data.

15.3 Environment membership link

A Person may be linked to an Environment membership. The Person detail page then exposes membership identity, role, active/inactive state and whether a local Environment password is configured.

A Person linked to an Environment membership cannot be deleted until the membership is unlinked.

Create a Person with a preferred name and ORCID, affiliate the Person with an Organization and a Group, then relate a Task to that Person using **With** . Open the Person and confirm that the Task appears in Related Work.

CHAPTER 16

Organizations and Groups

16.1 Organizations

Organizations model institutional structure such as universities, institutes, departments, companies or observatories. Organizations can themselves be contained in other Organizations.

The editor can expose fields such as full name, short name, website, email, phone, address, city, postal code, country, tags, symbol, dates and notes. Long-form Notes use a larger, clearly bounded writing surface so that editable text is visually distinct from the surrounding form.

16.2 Groups

Groups model teams, laboratories, committees, collaborations and working groups. A Group may be independent or contained by one or more compatible structural entities according to the current hierarchy rules.

16.3 Affiliated People

Organization and Group editors can manage affiliated People. Their Related Work pane is deliberately explicit: a Work item appears only when it has a relationship to the exact selected Organization or Group. Work is not included merely because it belongs to affiliated People, descendant Groups, descendant Organizations or another structural relation. A relationship copied onto a child Work item is stored on that child and therefore continues to count as a direct relationship.

> Create an Organization called *Institute*, a child Organization called *Department*, and a Group called *Lab* inside the Department. Affiliate a Person with the Lab. Relate a Task to the Lab and verify Related Work from the relevant structural views.

CHAPTER 17

Work Relationships

Work relationships attach professional context directly to Work items.

Available roles are:

- For;
- Requested by;
- With;
- Assigned to;
- Related to.

Each relationship points to a Person, Group or Organization.

In the Work editor, relationship entities are selected through the same native Testudo selector used elsewhere in the application. The selector separates People, Groups and Organizations and supports the existing searchable entity-selection workflow rather than presenting one generic Entity dropdown.

17.1 Inheritance by child items

A relationship may be marked **Inherited by child items**. Child Work can then inherit relevant context from its parent Task.

When changing the Parent Task in the full editor, Testudo may ask whether to use the new Parent's inherited relationships. Copying them replaces the current related People, Groups and Organizations with the Parent relationships marked for inheritance.

Create a parent Task related **For** an Organization and enable inheritance. Create a child Task and inspect its inherited context. Then move the child to another Parent Task with a different inherited relationship and test the **Copy Parent Relationships** / **Keep Current Relationships** decision.

CHAPTER 18

Related Work

People, Themes, Organizations and Groups share a Related Work presentation.

Related Work can contain Tasks, Notes and Activities. When related Work exists, the detail area can present the Work separately from the object's own inspector information.

Aggregation follows the relevant semantic structure:

- parent Themes can include descendant Theme Work;
- Organizations and Groups can aggregate from relevant descendants;
- duplicate Work is suppressed when several paths resolve to the same item.

Rows are navigable, so Related Work is also a bridge between the structural and Work domains.

Open a Person or Theme that has several related Work items. Select a related Task, then use Back. Verify that the original Person/Theme and its corresponding left/middle-column context are restored.

CHAPTER 19

Planned Start

19.1 Purpose

Planned Start is an optional date and time specifying when a Task is intended to become actionable. It is independent of Deadline, workflow status and the actual Started timestamp.

A Task with a future Planned Start can remain To Do. Reaching the planned date does not automatically change its status.

19.2 Subtasks

A Subtask inherits the availability constraints of its Parent Task and all ancestor Tasks.

The effective Planned Start is the latest applicable date and time in that hierarchy.

Tasks already In Progress or actually started remain available.

19.3 Today and Calendar

Today distinguishes work that is already actionable from work with future Planned Starts.

Calendar and Day workspace display explicit Planned Start occurrences. A Subtask inheriting only its Parent's Planned Start does not create an independent Calendar occurrence.

19.4 History

Creating, changing or removing Planned Start records a distinct historical event. This does not represent an actual Task start.

19.5 Compatibility

When Planned Start data is first saved, the Work Environment database transitions from schema 8 to schema 9.

This transition is retained even if Planned Start is later removed.

Testudo 0.3.0 cannot open schema 9 Environments.

Before using Planned Start in an Environment shared between Macs, update all relevant Testudo installations and preserve an independent backup of the Environment.

CHAPTER 20

Today

20.1 Today is a work summary, not only a schedule

The Today view combines several kinds of current-day relevance. In 0.1.7 the interface includes sections such as:

- Needs Attention;
- In Progress;
- Upcoming;
- Done Today;
- Discontinued Today;
- Today's Log.

The underlying Today collection considers active Work whose Created, Logged/Occurred, Started, Completed or Discontinued date is today.

20.2 Needs Attention

Needs Attention highlights urgent active items such as overdue deadlines, work due today and relevant reminders. Closed Task trees are excluded because archived Work is not part of the active workflow.

20.3 Upcoming

Upcoming provides near-future actionable context, such as deadlines and reminders.

Create one Task due today, start another Task today, complete a third Task today, discontinue a fourth Task today and create an Activity with today's Occurred time. Open **Today** and identify where each item appears. Then Close one terminal top-level Task and confirm the archived tree no longer contributes to the active Today workflow.

CHAPTER 21

Task Lists and Timeline

The sidebar exposes dedicated Task filters:

- All Tasks;
- To Do;
- In Progress;
- Completed;
- Discontinued.

These lists operate on active Work. Archived Task trees, including Tasks with unfinished child statuses, are excluded until reopened. Parent Task selectors for Tasks, Notes and Activities also exclude archived Tasks. Existing relationships inside the archived hierarchy remain preserved. Reopening restores the subtree to active views without changing its retained workflow statuses.

21.1 Timeline

Timeline focuses on chronological Work activity rather than a single workflow state. Activities and lifecycle dates help make long-running work understandable after the fact.

> Create four Tasks, one in each workflow status. Confirm that each appears in its corresponding sidebar filter. Close the Completed and Discontinued top-level Tasks and verify that they move to Archive and disappear from the active status filters. Reopen them and verify that they return to their retained status filters.

CHAPTER 22

History and the Work Log

22.1 Audit timestamps

The History section exposes key timestamps such as Created and Updated, plus type-specific lifecycle values such as Started, Completed, Discontinued or Occurred.

22.2 Log events

The chronological Log can represent:

- created;
- edited;
- status changed;
- scheduled;
- started;
- completed;
- discontinued;
- closed;
- reopened;
- activity logged;
- relationship added or removed;
- moved.

The Log is deliberately not only a mirror of the current state. Earlier lifecycle cycles remain meaningful historical records.

22.3 Deletion and history

Deleting a Work item is different from discontinuing or closing it. Deletion permanently removes the item and its descendants, plus associated relationships and history for those deleted Work items.

> If a Task existed and its outcome matters, prefer Completed, Discontinued and possibly Close. Use deletion for data that should genuinely be removed rather than preserved as history.

CHAPTER 23

Calendar Overview

23.1 What appears in Calendar

The Calendar combines several scheduling concepts:

- Calendar Events;
- Task deadlines;
- Note reminders.

The month view includes navigation for previous/next month and **Today** . Day summaries distinguish Events, Deadlines and Reminders.

The Calendar has a selected Day at all times. When Calendar is opened for the first time, the selected Day defaults to Today. Selecting another date changes that selection, and returning to Calendar preserves the selected date rather than unnecessarily resetting it to Today.

Selecting Calendar in the sidebar presents the currently selected Day as a first-class Day workspace in the right-hand detail pane.

23.2 Day workspace

The Day workspace combines several views of the selected date:

- **Schedule** – Calendar Events on that date;
- **Planned Work** – automatically selected actionable Tasks;
- **Due & Reminders** – Task deadlines and Note reminders belonging to the date;
- **Day History** – real Work-domain events that happened on the date;
- **App Log** – the recorded Testudo Work audit for the date.

Day pages participate in the same browser-style Back and Forward history as Tasks, Calendar Events, Themes and professional entities.

23.2.1 Planned Work

Planned Work is deterministic and derived from the current Work state. It considers Task hierarchy, deadlines, In Progress work and relationships to Calendar Events on the selected day.

Today may surface overdue work because it requires current attention. Future days do not repeatedly inherit Tasks whose deadlines have already passed merely because those Tasks remain To Do or In Progress.

Past dates do not reconstruct a fictional historical plan when no persisted plan snapshot exists.

23.2.2 Day History

Day History is a reverse-chronological record of real Work events, with the newest event first.

The Day History heading also displays a small secondary count. This number is the number of entries in the Day History collection for the selected date and therefore matches the entries presented below the heading.

It can include Tasks created as To Do, Tasks entering In Progress with their Started time, Completed and Discontin-

ued Tasks with their lifecycle time, Activities by Occurred time, and Notes by creation time.

Discontinued entries preserve the recorded reason and outcome where available.

23.2.3 App Log

App Log remains separate from Day History because it describes actions recorded by Testudo rather than the real-world Work chronology.

It is collapsed by default and can be expanded or collapsed with its disclosure control. When expanded, the newest recorded actions appear first.

23.3 Date navigation

Read-only dates and date-times throughout Testudo can open the corresponding Day page where appropriate.

Date navigation participates in normal detail-pane Back and Forward history. Editing controls such as date selectors remain editing controls rather than navigation links.

Toolbar actions include:

- Connect iCal Calendar;
- Connect Apple / iCloud Calendar;
- Sync connected external calendars;
- New Calendar Event.

23.4 Calendar sources

Current source kinds include:

- Testudo local calendar;
- iCal Subscription;
- Apple Calendar / iCloud.

For the same date, create a Task deadline, a Note reminder and a Calendar Event. Open that date in Calendar and confirm that all three appear but are identified by their different scheduling roles.

CHAPTER 24

Creating and Editing Calendar Events

24.1 New Calendar Event

A local Calendar Event can contain:

- Title;
- Calendar;
- All day flag;
- Start and End;
- source time zones for timed events;
- Location;
- Notes;
- Related Tasks;
- Related Themes.

The End cannot be earlier than Start.

Related Tasks are selected from the Task hierarchy, and related Themes are selected from the Theme hierarchy.

24.2 Calendar Event status

Calendar event statuses include Confirmed, Tentative and Cancelled where applicable to the event model.

24.3 External events

Events imported from external calendars are read-only for their external details. Testudo relationships can still be edited. The detail view explicitly states when an event is a read-only external calendar event.

Create a local Event called *Project review*, link it to a Task and a Theme, then open the Event and navigate through those links. Return with Back and verify the Calendar context is restored.

CHAPTER 25

Apple Calendar Integration

25.1 Connecting Apple Calendar

Choose **Connect Apple / iCloud Calendar** . Testudo requests macOS Calendar access and lists calendars available through Apple Calendar on the Mac.

The connection view states that imported events are read-only in Testudo and that only Testudo relationships can be edited.

The initial synchronisation window covers approximately one year in the past through two years in the future.

25.2 Permission and availability

If macOS permission is not granted, or no compatible calendars are available, the connection view reports the condition. Permission is controlled by macOS privacy settings.

> Connect a non-critical Apple Calendar, sync it and open one imported event. Confirm that event details are read-only while Related Work / Theme links remain manageable inside Testudo.

CHAPTER 26

iCal Subscriptions

Choose **Connect iCal Calendar** and provide the secret address in iCal format. An optional Calendar name can be entered.

Events imported from an iCal subscription are read-only.

The secret subscription address is treated as sensitive and is stored in macOS Keychain rather than in ordinary Testudo data.

The Keychain credential is local to the Mac on which it was entered. A shared `.testudoenv` package stores only the identifier used to locate that local credential; it does not contain the Secret iCal address itself.

When the same Work Environment is opened on another Mac, the Calendar and its imported events may therefore already be present while the local Secret address is absent. Use **Restore Secret Address** to provide the credential on that Mac. Restoration keeps the existing Calendar identity and updates the existing imported subscription rather than creating a second Calendar.

A missing local Keychain credential is treated as a machine-local condition and is not written into the shared Work Environment as a synchronization error.

Anyone with a secret subscription URL may be able to read the calendar feed. Do not place the URL in screenshots, Notes or shared documentation.

If you have a disposable iCal feed, connect it, trigger external-calendar synchronisation and inspect an event. Verify that the source behaves as read-only and that the feed URL is not shown as ordinary Work Environment data.

CHAPTER 27

Time Zones and Historical Dates

27.1 Absolute time and source zone

Testudo stores absolute timestamps as dates and, where relevant, stores a separate IANA time-zone identifier describing the zone in which the value was entered or recorded.

Time-zone-aware fields include, depending on type:

- Created and Updated;
- Scheduled;
- Deadline;
- Reminder;
- Started;
- Completed;
- Discontinued;
- Closed;
- Activity Occurred;
- Calendar Event Start and End.

The current device time zone is the normal display context, while the source zone helps preserve meaning across travel and historical entry.

27.2 Historical entry

Started, Discontinued and Activity occurrence times can describe events before the item was entered into Testudo, subject to logical validation rules.

> Create an Activity with a historical time and explicitly choose another IANA time zone. Save it, then reopen the detail and editor. Verify that the absolute moment and source-zone context remain consistent.

CHAPTER 28

Demo Environment

28.1 Purpose

Load Demo Environment creates a complete fictional data set intended for exploration. It includes People, Organizations, Groups, Themes, Tasks, sub-tasks, Notes, Activities, Calendar Events, affiliations, relationships, chronology, history and archived work.

Dates are generated relative to the date on which the Demo is created, so the environment remains useful over time.

28.2 Demo naming

Multiple Demo Environments can coexist. The generator uses names such as:

```
Demo
Demo-2
Demo-3
```

Available numbering can be reused after a Demo is removed.

28.3 Suggested tour

Use the Demo to inspect, in order:

1. Today;
2. one multi-level Task tree;
3. a Person with affiliations and Related Work;
4. a Theme with descendant-related Work;
5. Calendar;
6. Archive;
7. browser-style Back/Forward navigation across different object types.

> Before upgrading an important real Environment, create a fresh Demo and perform the tour above. It is a low-risk way to detect obvious UI or persistence regressions.

CHAPTER 29

Storage, Portability and Backups

29.1 Portable Environment package

The `.testudoenv` package is the principal portable unit. It contains the Environment's manifest, data, credentials and optional icon.

29.2 Storage compatibility in Testudo 0.3.0

Testudo 0.3.0 introduces explicit compatibility verification before supported Environment data is opened or written.

The verification checks the available Environment format information, database representation and Environment identity. It rejects unsupported formats, incompatible schema versions, inconsistent identities and malformed data rather than attempting to repair them silently.

The normal legacy snapshot storage engine supports compatible schema 8 databases. Database schema 7 requires a separate migration procedure and is not migrated automatically.

A future database schema is not automatically downgraded. Journal-based storage also requires a different storage engine and cannot be treated as a legacy snapshot database.

Supported older manifest metadata can be updated conservatively when verification permits. A missing manifest is not silently reconstructed from unverified data.

> If an Environment is reported as incompatible, preserve its complete original package and an independent backup. Do not manually change its schema number or remove files to bypass compatibility checks. A necessary migration should be performed separately with a verified procedure.

29.3 Safer startup and Environment switching

Startup verification avoids automatically rewriting an Environment database.

When registering or activating an Environment, Testudo validates the relevant format and identity before accepting it.

When switching between Environments, the application protects the current selection and in-memory state if saving pending data or updating the application registry fails.

A failed switch should therefore be investigated rather than bypassed by manually replacing registry or Environment files.

29.4 Environment credential-file safety

Each Work Environment maintains its own credential file, separate from the installation-level Testudo password.

Before a normal Environment credential update, Testudo verifies that the Environment database can be written safely and that the credential data belongs to the active Environment.

An existing credential file must be readable and use a supported schema. The update must also preserve existing

credential membership identities.

A missing credential file is not silently recreated by the normal password writer. Corrupt or incompatible credentials are preserved instead of being replaced with an empty set.

The first Administrator is initialized through the dedicated Environment creation workflow.

29.5 Administrator session requirement

An Environment Administrator must have an open Environment session to perform Administrator password changes.

Closing the Environment session removes this authorization even when the Environment remains selected and its remembered access information has not been deleted.

Password recovery is a separate workflow requiring its own verification.

29.6 Cloud-synchronised folders

An Environment can live in Dropbox, iCloud Drive, OneDrive or another filesystem provider visible to macOS. Testudo still works on the package as a file package.

Avoid simultaneous conflicting edits from several Macs. Wait for file synchronisation to complete before opening the same package elsewhere.

29.7 Backups

File synchronisation is not a substitute for independent backup. A deletion or corruption can itself be synchronised.

Recommended practice:

1. keep at least one independent backup of important Environments;
2. before upgrading between early versions, copy critical `.testudoenv` packages;
3. avoid manually editing JSON inside production Environment packages;
4. if using cloud storage, confirm synchronisation has completed before moving computers.

CHAPTER 30

Security and Privacy

30.1 Passwords

Application and Environment passwords are separate. Password verification uses salted password hashes rather than readable plaintext storage.

30.2 Security questions

Recovery answers are stored only as salted hashes. Administrators cannot view other users' recovery answers.

30.3 External calendar secrets

Secret iCal URLs use macOS Keychain integration rather than ordinary Environment data. The Secret value is machine-local; it is not transferred merely because a `.testudoenv` package is synchronized to another Mac. Since Testudo 0.1.9, the credential for an existing subscription can be restored without changing that Calendar's identity.

30.4 Sensitive portable files

A `.testudoenv` package may contain substantial professional context, including entities, relationships and history. Treat it as potentially sensitive when copying, sharing or backing it up.

CHAPTER 31

Deletion Semantics

Deletion is permanent and differs from lifecycle transitions.

Task / Note / Activity Deleting a Work item removes it and descendant Work, together with associated relationships and history for the deleted items.

Theme Deleting a Theme can also delete descendant Themes and Work assigned inside that Theme hierarchy according to the confirmation shown by the application.

Organization / Group Deleting the entity removes structural memberships and Work relationships; contained entities themselves are retained where the application confirmation states so.

Person Deletion removes the Person profile, structural memberships and Work relationships, but is blocked while the Person is linked to an Environment membership.

Calendar Event Deletion permanently removes the event from Testudo; an externally synchronised deletion can be queued for synchronisation where supported.

> Deletion scope differs by object type. Testudo's confirmation message is the authoritative description of what will be removed. For Tasks with meaningful history, prefer Completed, Discontinued and Archive instead of deletion.

CHAPTER 32

Recommended Working Practices

32.1 Separate plan, context and evidence

A useful Testudo pattern is:

- Task for what must be done;
- Note for stable context or reference material;
- Activity for what actually happened;
- Calendar Event for reserved or scheduled time.

32.2 Use Discontinued instead of pretending success

If work ended without successful completion, use Discontinued with a reason and optional outcome. This gives later readers a truthful explanation.

32.3 Close only when the top-level work is no longer part of active workflow

A Completed or Discontinued Task can remain visible in active status lists for as long as it is still operationally relevant. Close it when the whole top-level Task tree belongs in Archive.

32.4 Use Themes for concepts, not as a duplicate Task tree

Themes are semantic classification. Parent Task relationships should represent Work decomposition, while Theme hierarchy should represent conceptual grouping.

32.5 Model recurring professional context explicitly

If the same Person, lab, institution or committee repeatedly appears, create an entity and relationship rather than repeatedly typing names into free text.

32.6 Record outcomes as Activities

For long-running work, short Activities documenting important events provide a more useful record than continuously rewriting the Task description.

CHAPTER 33

Troubleshooting

33.1 An Environment requires migration

If Testudo reports that an Environment uses an older database schema, do not edit the schema version manually.

Testudo 0.3.0 does not automatically convert schema 7 Environment databases to schema 8. Preserve the original package and a backup, then use an explicitly verified migration process before opening it with the newer application.

33.2 Credentials are missing or incompatible

If a password update reports missing, corrupt, incompatible or mismatched Environment credentials, stop attempting to reset passwords.

Preserve the complete Environment package. Check that the correct Environment was selected and investigate the credential file before attempting recovery.

The ordinary password writer deliberately refuses to recreate a missing credential file or overwrite unreadable credentials.

33.3 An Administrator cannot change a password

Make sure the correct Work Environment is active and that you have opened an authenticated Environment session.

An Administrator membership alone is not sufficient when the Environment session has been closed.

33.4 The installer says the source checkout has local changes

The automatic installer deliberately refuses to overwrite local work. Inspect the checkout:

```
cd ~/Testudo
git status
```

Commit, stash or intentionally discard the local changes before using the automatic updater.

33.5 Apple developer tools are missing

Request Apple's Command Line Tools:

```
xcode-select --install
```

Then rerun the compatibility check or installer.

33.6 An Environment package was moved

If the registered path or security-scoped bookmark no longer points to the package, use **Open Existing Environment...** and select the package at its new location.

33.7 Environment login fails

Check:

- the correct Environment is active;
- the username is correct;
- the membership is Active;
- a password is actually configured for the membership;
- password recovery questions are configured if using recovery.

If recovery is unavailable, contact an Environment Administrator.

33.8 Apple Calendar does not appear

Check macOS Calendar permission, verify that the desired calendar is available in Apple Calendar, reconnect if necessary and run external-calendar synchronisation.

33.9 iCal synchronisation fails

Possible causes include an invalid or expired secret address, connectivity problems, changes at the calendar provider, or a Secret address that has not yet been stored in the current Mac's Keychain. If Testudo reports that this Mac does not have the Secret iCal address, use **Restore Secret Address** and enter the existing subscription URL. This restores the credential in place and should be preferred to disconnecting and recreating the Calendar when the existing Calendar identity and relationships must be preserved. If macOS asks for Keychain access after Testudo has been rebuilt, authorize the application and retry synchronization.

33.10 A Task cannot be Closed

Closing is available only for top-level Tasks whose status is Completed or Discontinued. Complete or Discontinue the Task first. Child Tasks are archived through their closed top-level ancestor.

33.11 A Discontinued status change appears unavailable in a shortcut

Discontinued requires a reason and date. Use the full Work editor so the required metadata can be entered and validated.

33.12 Archived Work is missing from active lists

This is expected. A Closed top-level Task and its subtree are removed from active Task, Today and related active-work queries. Open Archive or Reopen the root Task.

CHAPTER 34

Guided End-to-End Verification Lab

This chapter provides a complete hands-on test of the core Work model and interface in one disposable Environment.

34.1 Step 1: create the Environment

Create a new Environment named *Manual Verification*. Create its Administrator account and enter the Environment.

Expected result: a fresh Environment opens with no Work data other than any application defaults.

34.2 Step 2: create professional context

Create:

- Organization: *Example Institute*;
- Group: *Field Team*;
- Person: *Alex Example*;
- Theme: *Campaign 2027*.

Affiliate the Person with the Group and/or Organization as appropriate.

Expected result: each object has a read-only detail inspector and whole-object Edit action, and navigation links work between structural objects.

34.3 Step 3: create a top-level Task

Create Task *Prepare campaign* assigned to Theme *Campaign 2027*. Add a relationship **With** Alex Example and enable inheritance for child items. Set a future Deadline.

Expected result: the Task appears in To Do and in Related Work for the selected Theme/Person where appropriate.

34.4 Step 4: create children

Inside the Task create:

- sub-task *Check instruments*;
- Note *Shipping instructions*;
- Activity *Kick-off discussion* with an Occurred time.

Expected result: the Task's Contains section shows all three children and inherited relationships behave according to the relationship settings.

34.5 Step 5: exercise status and Today

Change *Check instruments* to In Progress. Confirm Started is populated or set manually. Create another Task due today.

Expected result: Today shows relevant active work in the appropriate Needs Attention / In Progress / Upcoming areas.

34.6 Step 6: exercise Discontinued

Create Task *Use legacy sensor*. Edit it to Discontinued, reason *Superseded*, outcome *New sensor selected*.

Expected result: the Task appears in Discontinued, its detail page shows the metadata, and the Log contains the lifecycle event.

34.7 Step 7: exercise Completed and Archive

Complete the top-level *Prepare campaign* Task, then choose Close.

Expected result:

- the complete subtree moves to Archive;
- the root still reports Status = Completed;
- active Task lists no longer include the tree;
- Close is present in history.

34.8 Step 8: navigation history

While viewing the archived Task, open a Person, then a Theme. Use Back twice and Forward twice.

Expected result: sidebar section, middle-column selection and right-pane content travel together through the history.

34.9 Step 9: Reopen

Return to the archived root and choose Reopen.

Expected result: the Task tree leaves Archive, returns to active views with Status = Completed, and the history contains both Closed and Reopened.

34.10 Step 10: calendar

Create a local Calendar Event, link it to the Task and Theme, and set a location. Also create a Note reminder and Task deadline on the same date.

Expected result: Calendar distinguishes Event, Reminder and Deadline while still allowing navigation to the related Work.

If all ten steps behave as described, you have exercised the current core Testudo Work model, Task lifecycle, Archive semantics, three-column navigation, semantic relationships, Today, Calendar and persistence-facing workflows.

CHAPTER 35

Technical Reference

35.1 Important file extensions

Extension	Purpose
`.testudoenv`	Portable Work Environment package.
`.testudouser`	Portable local Testudo profile information; does not contain Environment registry or passwords.
`.dmg`	Locally generated macOS installer disk image.

35.2 Source tree

```
Sources/Testudo/
|-- Auth/
|-- Demo/
|-- Models/
|-- Resources/
|-- Store/
|-- Time/
|-- Views/
`-- TestudoApp.swift
```

35.3 Build scripts

```
Scripts/build-app.sh
Scripts/build-release.sh
Scripts/build-local-dmg.sh
Scripts/build-manual.sh
Scripts/version.sh
```

35.4 Version identity

`Scripts/version.sh` is the build scripts' central version source. For this manual:

```
TESTUDO_VERSION="0.3.2"
TESTUDO_BUILD_NUMBER="18"
```

35.5 Task status persistence compatibility

The current user-visible Task status set excludes Closed. A legacy persistence-only Closed value remains decodable so Environment files written by Testudo 0.1.6 and earlier can be migrated to the separate `closedAt` archive lifecycle.

CHAPTER 36

Glossary

Activity A Work item recording something that happened, optionally with an Occurred timestamp.

Archive The view containing top-level Tasks that have been Closed, together with their retained subtrees.

Calendar Event A scheduled calendar-domain object, distinct from Activity.

Closed An archive lifecycle state for terminal top-level Tasks; not a user-facing Task workflow status.

Completed A terminal Task status representing successful completion.

Discontinued A terminal Task status representing work that ended without successful completion and carries a reason/date.

Entity A Person, Group or Organization.

Environment Administrator A membership with administrative privileges inside one Work Environment.

Environment membership An identity belonging to one Work Environment, separate from the local Testudo profile.

Group A laboratory, team, committee, collaboration or similar collective entity.

Local Testudo profile The installation-level professional profile of the Testudo user.

Note An informational Work item without Task status.

Organization An institutional entity that can participate in structural hierarchy.

Person An individual represented within a Work Environment.

Related Work Work associated with a Theme or professional entity directly or through supported descendant aggregation.

Reopened A lifecycle event that returns a Closed top-level Task from Archive while preserving its real workflow status.

Task An actionable Work item with one of the user-facing statuses To Do, In Progress, Completed or Discontinued.

Theme A hierarchical semantic classification independent of Work and entity hierarchies.

Work Environment An independent portable workspace represented by a `.testudoenv` package.

Work relationship A semantic link between Work and a Person, Group or Organization.

CHAPTER A

Command Reference

A.1 Check compatibility

```
curl -fsSL \
  https://raw.githubusercontent.com/foivoskar/Testudo/main/install.sh \
  | bash -s -- --check
```

A.2 Install

```
curl -fsSL \
  https://raw.githubusercontent.com/foivoskar/Testudo/main/install.sh \
  | bash
```

A.3 Clone manually

```
git clone https://github.com/foivoskar/Testudo.git
cd Testudo
```

A.4 Update source

```
git pull --ff-only
```

A.5 Build local installer

```
./Scripts/build-local-dmg.sh
```

A.6 Build application only

```
./Scripts/build-app.sh
```

A.7 Request Apple Command Line Tools

```
xcode-select --install
```

CHAPTER B

Manual Source and Compilation

The manual source lives under:

```
Documentation/TestudoManual/
```

The maintained source files are:

```
Testudo_manual.tex
testudomanual.cls
```

This edition deliberately keeps `testudomanual.cls` unchanged.

The normal repository-level build command is:

```
./Scripts/build-manual.sh
```

The script compiles the manual with XeLaTeX using the existing class file and writes the finished PDF at the repository root. A direct build remains possible:

```
cd Documentation/TestudoManual
latexmk -xelatex Testudo_manual.tex
```

The repository-level final copy is named:

```
Testudo_manual.pdf
```

CHAPTER C

Quick Acceptance Checklist

Use this checklist after installing a new Testudo build:

1. Local login works and Settings are accessible.
2. A Work Environment can be opened or created.
3. New Entry creates Task, Note and Activity correctly.
4. Task status filters show To Do, In Progress, Completed and Discontinued.
5. Discontinued requires a reason and preserves outcome metadata.
6. Only terminal top-level Tasks expose Close.
7. Archive preserves the actual Completed/Discontinued status.
8. Reopen returns the subtree to active views.
9. Back/Forward restores all three columns, including Archive destinations.
10. Closed Task trees remain absent from active **All Tasks** .
11. Day History shows a count matching the entries displayed for the selected day.
12. Command-P opens native printing for the right-hand detail content and long detail views paginate across multiple pages.
13. **File > Export...** creates a Beta semantic PDF from a supported detail destination.
14. Nested Task export offers depth values only up to the real maximum depth of the selected subtree.
15. Today reflects current-day Work lifecycle activity.
16. Calendar opens the currently selected Day in the detail pane and initially defaults to Today.
17. Calendar shows Events, deadlines and reminders.
18. External calendar events remain read-only except for Testudo relationships.
19. Person/Theme/Organization/Group Related Work navigation works.
20. A disposable Environment can be closed and reopened without data loss.

CHAPTER D

Early Release Notice

Testudo is under active development. Screens, labels and individual workflows may evolve between early versions. Before upgrading important data, keep an independent copy of critical `.testudoenv` packages.

License

Testudo is distributed under the MIT License. See the repository's `LICENSE` file for the complete text.

Author

Foivos Karakostas



Testudo User Manual
Manual version 2.3
October 2026